

E.3 / PAPER 1 / QUESTION 40

28 April 2022 / TZ2 / HL Paper 1 / original Q38 / 2222-6513

40. Samples of two radioactive nuclides X and Y are held in a container. The number of particles of X is half the number of particles of Y. The half-life of X is twice the half-life of Y.

What is the initial value of $\frac{\text{activity of radioisotope X}}{\text{activity of radioisotope Y}}$?

- A. $\frac{1}{4}$
- B. $\frac{1}{2}$
- C. 1
- D. 4